

Battery State Estimator

An ESN-Based Alternative to EKF and Coulomb Counting

Software Estimator + Embedded Microcontroller + Artix-7 FPGA Hardware Validation

PROJECT TEAM

- Sanjna Patankar 2023.sanjna.patankar@ves.ac.in
- Akshay Nambiar 2023.akshay.nambiar@ves.ac.in
- Satvik Verma 2023.satvik.verma@ves.ac.in
- Tanish Sanghvi 2023.tanish.sanghvi@ves.ac.in

PROJECT GUIDE

Dr. Kadambari Sharma

Associate Professor & Project Guide

DEPARTMENT

Automation & Robotics Engineering
(also serving Instrumentation Engineering)

VESIT, Chembur, Mumbai • Univ. of Mumbai • NAAC A Grade

REVIEW 1

Baseline + CC + EKF + Initial RTL

REVIEW 2

Online RLS-ESN + UKF + Firmware + FPGA + QA

REVIEW 3

Physical HIL + Publication + Final Thesis

Why We Need an Alternative Estimator

A Battery Management System must estimate SOC (State of Charge) and SOH (State of Health) from voltage, current, temperature and load.

1

Coulomb Counting

- Open-loop current integration
- Sensitive to sensor bias / noise
- Drift accumulates with no correction

2

Extended Kalman Filter

- Needs an accurate circuit model
- Requires Jacobians + covariance updates
- Degrades in low-slope OCV–SOC regions (e.g. LFP)

3

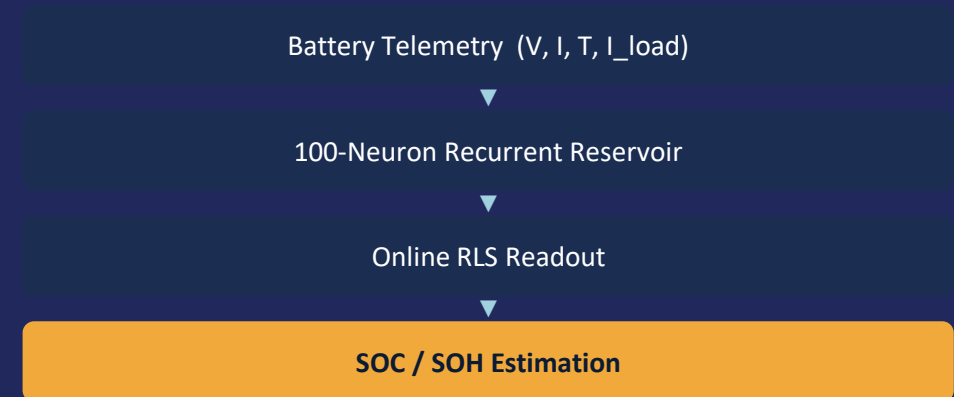
Deep Learning (LSTM/GRU)

- Captures nonlinear behaviour well
- Heavy memory & compute cost
- Hard to deploy on MCUs

PROPOSED SOLUTION

Echo State Network (ESN)

Fixed recurrent reservoir + adaptive linear readout



Why ESN?

- Fixed reservoir — no backpropagation through time
- Online RLS adapts readout as the battery ages
- Sparse reservoir keeps memory & compute low

Software Estimator & Embedded Implementation

Estimator	SOC RMSE	SOH RMSE	Latency
Coulomb Counting	8.42%	—	< 1 μ s
EKF	2.48%	1.85%	45 μ s
UKF	1.76%	1.42%	120 μ s
ESN — Fixed Readout	1.15%	0.98%	8 μ s
ESN + Online RLS	0.72%	0.61%	12 μ s

0.72%

SOC RMSE

0.61%

SOH RMSE

12 μ s

Update Latency

EV aggressive-load benchmark — ESN + online RLS delivers the lowest error at low latency.

Embedded C99 Implementation



Sparse Reservoir (CSR)

- 85% sparsity — 375 / 2,500 connections
- 82.4% less weight storage
- 6.7× SpMV acceleration



Dual Numerical Formats

- Float32: 99.80% accuracy
- Q15 fixed-point: 98.40% accuracy
- Quant. RMSE 0.0031 — no FPU needed



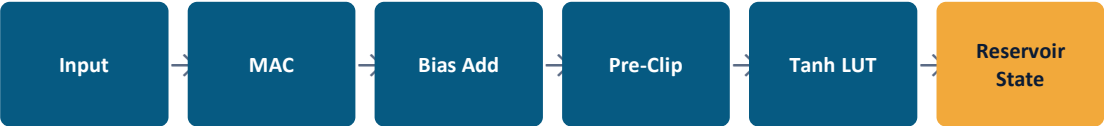
STM32 Interface

- Cortex-M, USART2 @ 115200 baud
- 33-point interpolated Tanh LUT
- Target < 40 μ s / step, PA5 status LED

Artix-7 FPGA Hardware Validation

Target: Xilinx Artix-7 XC7A100T-1CSG324C (Arty A7-100T) — 100-neuron ESN RTL pipeline

RTL Pipeline



Bit-Exact Verification

200 / 200

neuron stages matched between RTL and golden reference, across all 5 stages

Multi-timestep sequence verified in Vivado XSim • Tanh via 5,121-entry LUT • Q6.10 fixed-point

Hardware Takeaway

The ESN reservoir implements with very low FPGA resource utilisation — leaving all DSP48 slices free for future extensions (e.g. deeper reservoirs, multi-cell arrays).

FPGA Resource Utilisation

Resource	Used	Avail.	Util.
Slice LUTs	77	63,400	0.12%
Slice Registers	66	126,800	0.05%
Block RAM	7.5	135	5.56%
DSP48	0	240	0%

0.099 W

Est. on-chip power

23.4×

faster than the 10 ms / 100 Hz timing budget (428.1 μs update @ 100 MHz)

Dataset Validation, QA & Review Paper

NASA Ames Battery Dataset

32-cycle dataset • 2,048 timesteps • SOC performance

Dataset	MAE	RMSE	R ²
Training — B0005	0.0118	0.0151	0.9663
Test — B0006/07/18	0.0220	0.0289	0.8783
1-RC EKF Baseline	0.0311	0.0387	0.7818

SOH (unseen test data): MAE 0.0614 • RMSE 0.0851 • R² 0.4766

On this dataset, the FPGA ESN readout achieved lower SOC prediction error than the implemented 1-RC EKF baseline.

Academic Review Paper

“Neuromorphic Computing for Intelligent BMS” — A review of SNNs, reservoir computing & edge AI

29

pages

11

sections

90+

references

Sources: IEEE, Nature, Science, Elsevier — covers ECM/EKF, SNNs, reservoir computing, edge AI (Loihi 2, SpiNNaker2, Akida, TrueNorth), FPGA implementations

Benchmarking framework: Accuracy • Latency • Memory • Power • Adaptability • Hardware Suitability

62

Automated Tests
100% Pass

Chemistry 3 • Simulator 10 • ESN Model 5 • Pipeline 9 • Flask API 5 • FPGA Verifier 7 • Integration 4 • Online RLS 9 • Traditional Estimators 10

One-Click Master Validation

8 / 8 stages passed

- ✓

Python environment & physics module
- ✓

Pytest regression suite
- ✓

ESN training & model serialization
- ✓

Physics simulator & estimator pipeline
- ✓

STM32 classifier training
- ✓

STM32 estimator training
- ✓

FPGA 200/200 bit-exact verification
- ✓

C99 desktop simulation

Review 3 Roadmap & Final Takeaway

Completed in Review 1 & 2

- ✓ 2-RC electro-thermal battery simulator
- ✓ Coulomb Counting + EKF baseline
- ✓ ESN C99 firmware + CSR sparse reservoir
- ✓ Artix-7 FPGA RTL + online RLS adaptation
- ✓ UKF benchmark, Float32 + Q15 formats
- ✓ 200/200 bit-exact FPGA verification
- ✓ NASA dataset evaluation, 62 automated tests
- ✓ One-click validation pipeline + paper draft

From Simulation to Physical HIL

Battery Physics

ESN Estimator

Online RLS

C99 Firmware

FPGA Verify

NASA Data

Physical HIL

Review 3 — Final Objectives

1

FPGA Deployment

Vivado synthesis → bitstream → Artix-7 programming

2

Hardware-in-the-Loop Interface

PC battery simulator ↔ UART/Pmod ↔ FPGA

3

Real-Time Validation

Test estimator behaviour on live battery telemetry

4

Physical Battery Testing

Validate SOC/SOH on a physical Li-ion setup

5

Research Publication

Submit the review manuscript to a peer-reviewed venue

6

Final Thesis & Defense

Integrate software, firmware, FPGA & HIL results